

**Understanding Server Components** Overview of server hardware: CPU, RAM, Storage, and Network Interface Cards (NICs). Practical (2 hours): Assemble a virtual server environment (using virtualization tools like VirtualBox or VMware). Identify and explore hardware configurations.

### **Tasks**

1. Install VirtualBox or VMware Workstation Player on your computer and create a new virtual machine named 'MyServerLab' with default settings.

**Answer :-** To install a virtualization software and create a new virtual machine named MyServerLab.

#### **Step 1: Install VMware**

1. Download and install VMware Workstation Player or VMware Workstation.
2. Open VMware after installation.

#### **Step 2: Create a New Virtual Machine**

1. Click Create a New Virtual Machine.
2. Select Typical (Recommended).
3. Click Next.

#### **Step 3: Select the Operating System**

If you have an Ubuntu/Linux ISO file:

- Select Installer disc image file (ISO).
- Browse and select the Ubuntu ISO file.

If you only need to create the VM for this practical and will install Linux later:

- Select I will install the operating system later.

Click Next.

#### **Step 4: Set the Virtual Machine Name**

Enter: MyServerLab

Choose a suitable location to save the virtual machine.

Click Next.

### **Step 5: Configure the Hard Disk**

Set the disk size according to the assignment. If 40 GB is required:

Maximum disk size: 40 GB

Select: Store virtual disk as a single file

Click Next.

### **Step 6: Complete the Setup**

Review the settings and click Finish.

The new virtual machine will appear in VMware as: MyServerLab

### **Step 7: Verify the Virtual Machine**

Select MyServerLab in VMware and check that the VM has been created successfully.

Take a screenshot showing:

VMware window

VM name: MyServerLab

Virtual machine settings

### **Result**

A new virtual machine named MyServerLab was successfully created using VMware with the required/default settings.

**2.** Within your 'MyServerLab' virtual machine, configure the hardware settings to assign 2 CPU cores, 4GB RAM, and a 40GB virtual hard disk.<br><br><em><strong>Hint:</strong> Use the 'Settings' or 'Edit virtual machine settings' option before starting the VM.</em>

### **Answer :- Configure CPU, RAM and Hard Disk**

- Shut down the MyServerLab virtual machine.
- Open VMware and select MyServerLab.
- Click Edit virtual machine settings.
- Select Memory and set it to 4 GB.

- Select Processors and set the number of processor cores to 2.
- Select Hard Disk and check/set the virtual disk size to 40 GB.
- Click OK to save the settings.

**Result:** The MyServerLab VM was configured with 2 CPU cores, 4 GB RAM, and a 40 GB virtual hard disk.

**3.** Add a second virtual network interface card (NIC) to your 'MyServerLab' VM and set one NIC to NAT and the other to Bridged mode.  
  
**Hint:** This simulates how real servers can connect to multiple networks.

**Answer :- Configure Two Network Interfaces**

- Shut down the MyServerLab VM.
- Open VMware → MyServerLab → Edit virtual machine settings.
- Select Network Adapter and set the first adapter to NAT.
- Click Add, select Network Adapter, and click Finish.
- Select the newly added second adapter and set it to Bridged.
- Click OK.
- Final configuration:

Network Adapter 1 – NAT

Network Adapter 2 – Bridged

**Result:** Two network interfaces were successfully configured. One uses NAT and the other uses Bridged networking.

**4.** Boot your virtual machine and use the system information tool (like 'System Information' on Windows or 'lshw' on Linux) to list the detected CPU, RAM, storage, and network interfaces. Take a screenshot of the output.

**Answer :- Check CPU, RAM, Storage and Network**

- Start the MyServerLab VM and open the Ubuntu/Linux terminal.
- Check CPU :- `lscpu`

```
Oct 1 10:15
diksha@diksha-VMware-Virtual-Platform: ~
diksha@diksha-VMware-Virtual-Platform:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 45 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-1334U
CPU family: 6
Model: 186
Thread(s) per core: 1
Core(s) per socket: 1
Socket(s): 2
Stepping: 3
BogoMIPS: 4992.00
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon
rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pnt pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f1
6c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd lbrs tlbpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx snap clflushopt clw
b sha_ni xsaveopt xsavec xgetbv1 xsavec avx_vnni arat unip gfnit vaes vpclmulqdq rdpid movdiri movdir64b frsm md_clear serialtize flush_lid arch_capabilities

Virtualization features:
Hypervisor vendor: VMware
Virtualization type: full
Caches (sum of all):
L1d: 96 KiB (2 instances)
L1i: 64 KiB (2 instances)
L2: 2.5 MiB (2 instances)
L3: 24 MiB (2 instances)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0,1
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit: Not affected
L1tf: Mitigation; PTE Inversion
Mds: Mitigation; Clear CPU buffers; SMT Host state unknown
Meltdown: Mitigation; PTI
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Vulnerable: No microcode
Retbleed: Mitigation; IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Speculative store bypass: Not affected; Speculative Store Bypass disabled via prctl
```

- Check RAM :- free -h

```
diksha@diksha-VMware-Virtual-Platform:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.8Gi       1.3Gi       833Mi         42Mi         1.9Gi         2.4Gi
Swap:          3.6Gi           0B         3.6Gi
```

- Check Storage :- lsblk

```
diksha@diksha-VMware-Virtual-Platform:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0    4K  1 loop /snap/bare/5
loop1        7:1      0   74M  1 loop /snap/core22/2292
loop2        7:2      0 251.7M  1 loop /snap/firefox/7766
loop3        7:3      0  66.8M  1 loop /snap/core24/1643
loop4        7:4      0  66.8M  1 loop /snap/core24/2124
loop5        7:5      0  18.5M  1 loop /snap/firmware-updater/210
loop6        7:6      0  16.5M  1 loop /snap/firmware-updater/226
loop7        7:7      0 531.4M  1 loop /snap/gnome-42-2204/247
loop8        7:8      0  402M  1 loop /snap/mesa-2404/1839
loop9        7:9      0 614.5M  1 loop /snap/gnome-46-2404/164
loop10       7:10     0 531.5M  1 loop /snap/gnome-42-2204/263
loop11       7:11     0  91.7M  1 loop /snap/gtk-common-themes/1535
loop12       7:12     0  10.8M  1 loop /snap/snap-store/1270
loop13       7:13     0  50.1M  1 loop /snap/snapd/27591
loop14       7:14     0  11.8M  1 loop /snap/snap-store/1390
loop15       7:15     0   576K  1 loop /snap/snapd-desktop-integration/343
loop16       7:16     0   828K  1 loop /snap/snapd-desktop-integration/391
loop17       7:17     0   50.3M  1 loop /snap/snapd/27738
sda          8:0      0   20G  0 disk
├─sda1       8:1      0    1M  0 part
└─sda2       8:2      0   20G  0 part /
sdb          8:16     0    5G  0 disk
└─md127     9:127    0    5G  0 raid1 /mnt/raid1
sr0         11:0     1 1024M  0 rom
```

- Check network interfaces: ip addr

```
diksha@diksha-VMware-Virtual-Platform:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:03:df:19 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.233.130/24 brd 192.168.233.255 scope global dynamic noprefixroute ens33
        valid_lft 1505sec preferred_lft 1505sec
    inet6 fe80::20c:29ff:fe03:df19/64 scope link
        valid_lft forever preferred_lft forever
diksha@diksha-VMware-Virtual-Platform:~$
```

- sudo lshw -short

```
Oct 1 10:20
diksha@diksha-VMware-Virtual-Platform:~$ sudo lshw -short
[sudo] password for diksha:
H/W path          Device          Class          Description
-----
/0                 system          VMware Virtual Platform
/0/0               bus             440BX Desktop Reference Platform
/0/0               memory          86KiB BIOS
/0/1               processor       13th Gen Intel(R) Core(TM) i5-1334U
/0/1/0             memory          16KiB L1 cache
/0/1/1             memory          16KiB L1 cache
/0/2               processor       13th Gen Intel(R) Core(TM) i5-1334U
/0/5               processor       CPU
/0/5/95            memory          16KiB L1 cache
/0/6               processor       CPU
/0/6/96            memory          16KiB L1 cache
/0/7               processor       CPU
/0/7/97            memory          16KiB L1 cache
/0/8               processor       CPU
/0/8/98            memory          16KiB L1 cache
/0/9               processor       CPU
/0/9/99            memory          16KiB L1 cache
/0/a               processor       CPU
/0/a/9a            memory          16KiB L1 cache
/0/b               processor       CPU
/0/b/9b            memory          16KiB L1 cache
/0/c               processor       CPU
/0/c/9c            memory          16KiB L1 cache
/0/d               processor       CPU
/0/d/9d            memory          16KiB L1 cache
/0/e               processor       CPU
/0/e/9e            memory          16KiB L1 cache
/0/f               processor       CPU
/0/f/9f            memory          16KiB L1 cache
/0/10              processor       CPU
/0/10/a0           memory          16KiB L1 cache
/0/11              processor       CPU
```

This displays detected hardware such as CPU, RAM, storage, and network interfaces.

**5.** Change the RAM allocation for your virtual machine from 4GB to 2GB, restart the VM, and observe any performance changes or warning messages. Write a short note on how RAM size impacts server performance.

**Answer :- Change RAM from 4 GB to 2 GB**

- Shut down the MyServerLab VM.
- Open VMware and select MyServerLab.
- Click Edit virtual machine settings.
- Select Memory.
- Change the RAM from 4 GB to 2 GB.
- Click OK.
- Start the VM.
- Observe the system performance.

**Check the available RAM using:**

- **free -h**